

Learn HTML

Lesson 4: Images and Media

Images and Media – Study Notes

Key Concepts

1. **`<img>` element** – embeds raster images and requires a `src` and `alt` attribute.
2. **Alternative text (`alt`)** – provides a textual description for accessibility, SEO, and when the image cannot be displayed.
3. **Audio and video elements** – `<audio>` and `<video>` embed sound and moving picture content, with optional controls and fallback text.
4. **Responsive media** – using `width`, `height`, `srcset`, `sizes`, and CSS to make images adapt to different screens.
5. **Media file formats** – common image (`.jpg`, `.png`, `.gif`, `.svg`) and audio/video (`.mp3`, `.wav`, `.ogg`, `.mp4`, `.webm`) types and when to choose each.

Summary

Embedding visual and auditory content enriches web pages, but it must be done with both usability and accessibility in mind. The `<img>` tag is the simplest way to place a picture on a page; it points to an image file via the `src` attribute and must always include an `alt` attribute that describes the image for screen readers and for cases where the image fails to load.

Audio and video are added with the `<audio>` and `<video>` elements. Both support a `controls` attribute that automatically creates a play/pause bar, and they allow multiple source files (`<source>`) so browsers can choose a format they understand. Adding fallback text inside the tags gives users without media support a clue about the missing content.

Responsive design techniques—such as setting `max-width: 100%` in CSS, using `srcset`/`sizes`, or the `picture` element for art direction—ensure that media looks good on phones, tablets, and desktops without unnecessary bandwidth consumption. Selecting the right file format (e.g., SVG for vector graphics, WebP for efficient raster images, MP4 for video) further improves performance and accessibility.

Important Points

- **`<img>` syntax**: ``.
- **`alt` text rules**:
 - Describe the image's purpose, not its file name.
 - Keep it concise ("H125 characters").
 - Use empty `alt=""` for decorative images.
- **Responsive images**:
 - `srcset="small.jpg 600w, medium.jpg 1200w, large.jpg 1800w"`

- `sizes="(max-width: 600px) 100vw, 50vw"`
- Or wrap in `<picture>` with `<source media="(min-width: 800px)" srcset="large.webp">`.
 - **Audio**: `<audio controls> <source src="song.mp3" type="audio/mpeg"> <source src="song.ogg" type="audio/ogg"> Your browser does not support the audio element. </audio>`
 - **Video**: `<video controls width="640" height="360"> <source src="movie.mp4" type="video/mp4"> <source src="movie.webm" type="video/webm"> Sorry, your browser can't play this video. </video>`
 - **Common formats**:
 - Images: JPEG (photos), PNG (transparent graphics), GIF (simple animation), SVG (scalable vectors), WebP/AVIF (modern, high compression).
 - Audio: MP3 (most compatible), OGG (open source), WAV (lossless, large).
 - Video: MP4 (H.264, universal), WebM (VP8/VP9, open), OGG (Theora, less common).
 - **Accessibility**: Always provide `alt` text, captions (`<track kind="captions">`) for video, and transcripts for audio when possible.

Examples

1. Responsive Image with `srcset`

```

<<html
<!-- HTML -->


```

Explanation

- `src` is the fallback for older browsers.
- `srcset` lists three versions of the same image with their intrinsic widths.
- `sizes` tells the browser how much viewport width the image will occupy at different breakpoints, letting it pick the smallest adequate file.

2. Embedding a Video with Controls and Captions

```

<<html
<video controls width="640" height="360" poster="thumb.jpg">
  <source src="travel.mp4" type="video/mp4">
  <source src="travel.webm" type="video/webm">
  <!-- Caption file (WebVTT) -->
  <track src="travel-captions.vtt" kind="captions" srclang="en" label="English captions">
  Your browser does not support the video tag.

```

</video>

****Explanation****

- ``controls`` adds the play/pause UI.
- ``poster`` shows a placeholder image before playback starts.
- Two `<source>` elements ensure broader browser support.
- The `<track>` element provides closed captions for accessibility.

Quick Review

1. What two attributes are **required** on every `<img>` element, and why?
2. When should you use an empty `alt`` attribute (`alt=""`)?
3. Name one advantage of providing multiple `<source>` elements inside `<audio>` or `<video>`.
4. How does the `sizes`` attribute work together with `srcset`` to deliver responsive images?
5. List two modern image formats that offer better compression than JPEG or PNG.

Further Reading

- ****Responsive Images**** – MDN Web Docs: `<picture>`, `srcset``, and `sizes``.
- ****Web Accessibility**** – WCAG guidelines for alt text, captions, and transcripts.
- ****Media Formats**** – Comparison of JPEG, WebP, AVIF; MP3 vs. OGG vs. AAC; MP4 vs. WebM.
- ****Performance Optimization**** – Lazy loading images (`loading="lazy"`), media streaming, and using a CDN.
- ****SVG Basics**** – Inline SVG vs. external files, accessibility with `<title>` and `<desc>`.

